Chapter 19
Volatility Smiles and Surfaces

Five chapters have deferred to this one. Section 11.3 derived conditions a whole option chain must satisfy and said they would be used here. Section 12.8 built a digital as the limit of a call spread and left the next derivative, the butterfly, which Section 13.3 met as a state price, to be read off the market here. Section 15.11 measured implied volatilities ranging from 56.9% to 33.2% across six strikes on a single stock on a single day, and observed that the model requires all six to be identical. Section 16.4 gave the FX market's quoting conventions, and Section 18.10 warned that the Greeks inherit the model's error.

The subject of this chapter is that error: the model requires one volatility, and the market's prices imply a different one at every strike. It is large, stable, well documented, and *informative*. The pattern of implied volatilities across strikes and maturities is not noise around a correct model. It is the market's statement about the distribution of the underlying, written in the only language the market has agreed on.

Sections 19.1 and 19.2 describe the two shapes the smile takes and what they say about the distribution. Section 19.3 introduces volatility indices and variance swaps, Section 19.4 the models that fit a smile, and Section 19.5 the constraints a surface must obey. Sections 19.6 to 19.10 cover the term structure, index against single-stock smiles, trading, the smile's causes and the volatility risk premium; Sections 19.11 and 19.12 cover how the smile moves and how the surface is used.

19.1 TWO SHAPES

An implied volatility is not observed. It is Section 15.7's construction: the σ that makes Chapter 15's formula reproduce a price the market actually traded at, so each one is a single traded option inverted. Plot those numbers against strike for a fixed maturity and one of two characteristic shapes appears. That curve is a **smile**, and the same option chain has one for every listed maturity. Taken together they define a function of two variables, one implied volatility for each strike K and maturity T, and that function is the **volatility surface**. It is what a desk quotes, marks and hedges from. Put–call parity makes a call and a put at the same strike and maturity imply the same volatility (Section 15.11), so a surface carries one number per strike, read from the out-of-the-money side, where the bid–ask spread is tighter.

Equities: a skew. Figure 19.1 shows the shape that has characterized index and single-stock options since the crash of October 1987. Implied volatility falls as the strike rises, so that, measured in implied volatility, low-strike puts are expensive and high-strike calls cheap relative to at-the-money options; only at short maturities does the far call wing (the curve's high-strike end) turn up again. The pattern is called the **skew** or "smirk" rather than a smile because it is not symmetric.

Currencies: a smile. Section 16.4 gave the FX shape: implied volatility is lowest near the money and rises in *both* wings, which is the symmetric curve the word "smile" describes. It is tilted (the risk reversal quoted there measures by how much), but the tilt changes sign as sentiment moves, whereas equity skew does not; only in pairs with a known crash direction, an

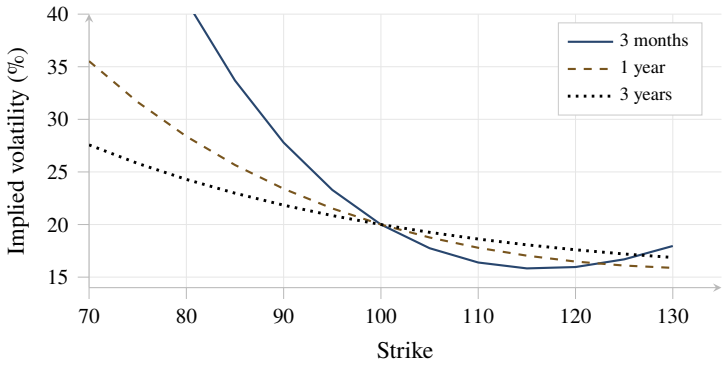


Figure 19.1: Equity implied volatility against strike, at three maturities. $S_0 = 100$; illustrative.

emerging-market currency or a carry trade's funding currency, does it keep one sign for years.

The difference is structural, and Section 19.9 explains it. An exchange rate is a ratio of two currencies, and a crash in one is a spike in the other, so the distribution has no preferred direction. An equity is a levered claim on a firm: it can be wiped out in a crash, but it has no matching way to multiply in value just as suddenly.

The term structure. Figure 19.1 also shows that the skew flattens as maturity lengthens. Measure the skew by its slope, the fall in implied volatility, in volatility points, for each 10 points of strike between strikes of 90 and 110: the slope falls from 5.7 at three months to 2.8 at one year to 1.6 at three years.

Most of that is the unit of measurement: the same 20 points of strike are about two standard deviations at three months and little more than half of one at three years, as the next subsection measures. The central limit theorem of Section 13.9 would flatten it faster than the change of unit alone, since independent shocks (jumps above all) average out over a long horizon, which is why jump models generate skew mainly at the short end (Sections 21.5 and 21.7). Volatility shocks are persistent and do not average out so quickly, and they keep the long-dated skew alive.

Measuring moneyness

The axis of Figure 19.1 is the strike, which is what the contract names and the worst of the available coordinates for comparing one option with another. Six are in use, set out in Table 19.1, and they are not interchangeable.

They are one axis in different units, and three things follow.

The coordinate decides what "the same option" means from one day to the next. Section 19.11's two rules for how the smile moves with the spot, sticky strike and sticky delta, are a disagreement about which coordinate the smile stands still in, not about today's prices, and (19.4) is the cost of that disagreement to a hedger.

A fixed strike is a different option at every maturity. At 20% volatility the 90 strike of Figure 19.1 is 1.05 standard deviations below the money at three months, 0.53 at one year and 0.30 at three, because $\sigma\sqrt{T}$ shrinks as expiration approaches. Measured per unit of standardized moneyness rather than per point of strike, the figure's three curves have slopes of 5.68, 5.59 and 5.55 volatility points per standard deviation: they are one curve in that coordinate. Real surfaces flatten more slowly than that: Section 21.7 measures the decay as $T^{-0.4}$ against the $T^{-1/2}$ a curve stationary in standardized moneyness would give, so in that coordinate the skew slowly *steepens* with maturity.

Delta is the only one a trader can act on directly, since it is the hedge ratio, already scaled by volatility and time. That is why FX quotes in it. It is also the circular coordinate: delta

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Table 19.1: The coordinates a surface is quoted in.

Measure	Definition	What it is for
Strike	K	The contract term, and how an exchange lists the option
Spot moneyness	K/S	Section 19.11's sticky-delta rule; the columns of Table 19.3
Forward moneyness	K/F	Removes carry, so at the money means $K = F$
Log-moneyness	$\ln(K/F)$	Scale-free: an underlying at 40 and one at 4,000 report the same skew for the same shape, which is why Chapter 21 measures in it
Standardized moneyness	$\ln(K/F)/(\sigma\sqrt{T})$	Distance from the money in standard deviations; comparable across maturities, as delta is
Delta	$N(d_1)$	The FX quoting convention of Section 16.4 (spot or forward delta, premium-adjusted in some pairs), already rescaled by volatility and time

depends on the volatility being quoted, so a 25-delta option's strike and its implied volatility depend on each other and have to be solved for together.

Shapes across asset classes

Table 19.2 sets out the surface's shape in each market; the differences are systematic.

Table 19.2: Characteristic surface shapes and quoting conventions.

Market	Shape	Why
Equity index	Steep skew	Leverage, crash risk, one-sided protection demand
Single stock	Milder skew	Same forces, plus takeover possibility lifting the right wing
Currency	Smile, tilt varies	Ratio of two economies; no preferred direction
Commodity	Often *reverse* skew	Supply shocks push prices up, not down
Interest rates	Skew, quoted in normal volatility	Rates can be negative, so lognormal fails outright

Commodities frequently skew the other way. A grain, natural-gas or power price is cushioned on the way down by production cuts and unbounded above when supply fails, so the fat tail is on the *upside*; crude oil and base metals, driven more by demand, often carry a put skew instead. Section 5.9's negative oil settlement does not undercut the cushioning argument: it required storage to run out.

Rates are quoted in normal volatility: the standard deviation of the rate's change measured in basis points, rather than of its percentage change. Chapter 29 explains why: once policy rates went negative, a lognormal model assigning zero probability to $r < 0$ became untenable, and the market moved to quoting basis-point volatility.

19.2 WHAT THE SMILE SAYS ABOUT THE DISTRIBUTION

Section 12.8 built a digital as the limit of a tight call spread, priced at $-\partial c/\partial K$. Section 13.3 pushed the same idea one derivative further, to a state price. An infinitesimal butterfly centered at K (the call spread with strikes $K - \varepsilon$ and K bought against the one with strikes K and $K + \varepsilon$ sold, scaled by $1/\varepsilon^2$) pays only if the underlying finishes at K, so its price is the state price for that outcome, and

$$\boxed{g(K) = e^{rT}\,\frac{\partial^2 c}{\partial K^2}} \tag{19.1}$$

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where g is the risk-neutral density of S_T and c is the market call price as a function of strike, so the derivatives run along the smile. This is the Breeden–Litzenberger result, and it means a complete set of option prices contains the market's entire probability distribution for the underlying. [TN19] derives it in two differentiations, shows why nonnegativity of the density is the convexity condition of Section 11.3, and sets out what changes when the chain is finite. Applied to a smile, not a single volatility, it makes the chapter's central point visible.

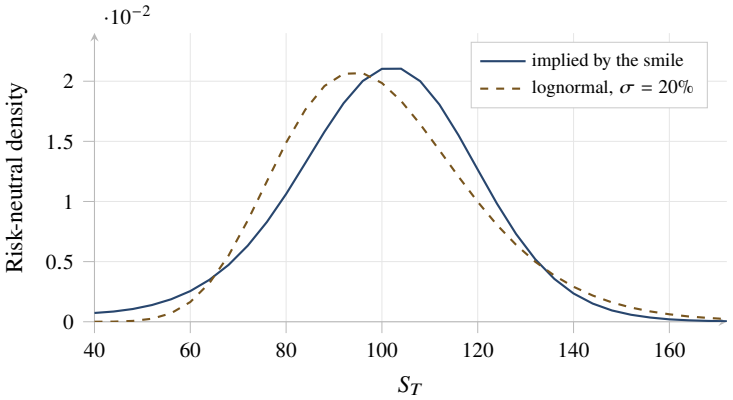


Figure 19.2: Density implied by a one-year equity skew, against the lognormal it would be if the model held. $S_0 = 100$, $r = 0$.

Example 19.1 Reading the tails

Figure 19.2 plots the density obtained from (19.1) applied to the one-year skew

$$\sigma(K) = 0.20 - 0.15 \ln(K/100) + 0.15 \ln^2(K/100)$$

the shape of Figure 19.1 with a milder left wing, against the lognormal density that a constant 20% volatility would produce.

Each probability is read off as a digital, $\Pr(S_T < K) = 1 + e^{rT}\, dc/dK$ with $r = 0$, where dc/dK is the call price's slope along the smile; this is the density integrated from zero:

	Implied by smile	Lognormal
$\Pr(S_T < 60)$	3.0%	0.7%
$\Pr(S_T < 70)$	6.9%	4.6%
$\Pr(S_T < 80)$	14.8%	15.5%
$\Pr(S_T > 130)$	6.0%	7.9%

The market assigns more than four times as much probability as the lognormal does to a 40% decline and 1.5 times as much to a 30% decline, and less probability to a 30% rise. Between the tails the implied density runs below the lognormal from about 65 to 97 and above it near the center, which is why the two columns nearly agree at 80: the distribution is peaked, with mass drawn from the left shoulder into the center and the far tail.

Expensive low-strike puts and cheap high-strike calls are not a pricing anomaly: they are what a fat left tail and a thin right tail look like when translated into Black–Scholes volatilities.

One caveat matters for practice. A tail probability at a quoted strike is the digital $1 + e^{rT}\, dc/dK$, so it depends only on the smile's level and slope there. Below the lowest traded strike the smile must be extrapolated, but the probability of finishing below that strike is still fixed: it is a digital put, which the put spread between the two lowest traded strikes approximates. The extension decides how that probability is spread over lower prices, and hence anything integrated over the tail; even that requires it to be free of arbitrage. Extended as a quadratic, this smile is not: the density turns negative below a strike of about 6, and integrated upward from that point it totals 1.06 rather than

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1.00, a 6% error created entirely outside the observable range.

The same calculation on real quotes

Figure 19.2 used a smooth parameterized smile. The point of (19.1) is that it needs no such thing: six quoted prices and arithmetic are enough, as Example 19.2 shows.

Example 19.2 Extracting a distribution from six quotes

Return to the option chain of Table 10.2: a stock at \$48.30, 45 days to expiration, six strikes \$2.50 apart. Section 15.11 solved each call's midquote (the bid–ask midpoint) for implied volatility:

Strike	Call mid	Implied vol	$C_{i-1} - 2C_i + C_{i+1}$	Density $g(K)$
42.50	6.650	56.9%	—	—
45.00	4.425	46.0%	0.4700	0.0757
47.50	2.670	39.7%	0.5600	0.0902
50.00	1.475	36.7%	0.4300	0.0692
52.50	0.710	34.5%	0.3600	0.0580
55.00	0.305	33.2%	—	—

The fourth column is the price of a butterfly with wings \$2.50 wide. Every entry is positive, so the chain passes Section 11.3's convexity test and admits no butterfly arbitrage. The fifth column divides by the wing width squared and compounds forward, which is (19.1) in difference form: $g(K) = e^{rT}(C_{i-1} - 2C_i + C_{i+1})/h^2$ with $h = 2.50$, $r = 5\%$ and $T = 45/365$.

Multiplying each density by \$2.50 gives, approximately, the probability the market assigns to each bucket, the \$2.50 band around a strike: 18.9%, 22.5%, 17.3%, 14.5%. The four together cover \$43.75 to \$53.75 and carry 73.3% of the probability.

The tails come from the outer call spreads. A call spread's price, compounded forward and divided by its width, approximates the digital at the midpoint of its strikes, so $\Pr(S_T > 43.75) \approx e^{rT}(6.650 - 4.425)/2.50 = 89.6\%$ and $\Pr(S_T > 53.75) \approx e^{rT}(0.710 - 0.305)/2.50 = 16.3\%$. The left tail is the complement of the first: $\Pr(S_T < 43.75) = 10.4\%$.

Why the total is exactly 100%. 10.4% + 73.3% + 16.3% = 100.0%, but the total is an *identity*, not a test: the middle differences telescope, leaving only the two outer call spreads, which are precisely the tail digitals, so *any* six numbers would sum to 100%.

A lognormal at the at-the-money implied volatility of 39.7%, centered on Example 11.4's forward of \$47.60, puts only 53.0% of its mass in that same central band. The lognormal that *does* put 73.3% there has $\sigma = 25.6\%$.

So the market is pricing the middle of the distribution as if volatility were about 26%, while the strike 12% below spot is quoted near 57%: the distribution is peaked in the center, and the little mass below \$43.75 must lie far enough out to justify that price. More mass in the center and in the far tails than a normal distribution carries is called **leptokurtosis**, and this is what it looks like when you measure it rather than assume it.

Two limits apply. The calls are American and the stock pays a dividend before expiration, so the deep-in-the-money \$42.50 quote, the one the left tail is read from, carries early-exercise premium a European call would not. Every \$0.10 of it lowers the measured $\Pr(S_T < 43.75)$ by about 4 points, so the 10.4% understates the European tail and the 57% overstates the wing. And the whole left tail rests on that one call spread: it fixes the 10.4%, but six strikes spanning −12% to +14% of spot cannot say how that 10.4% is distributed below them, as Example 19.1 warned. A chain tells you a tail's mass, and through the lowest put its mean, but never its shape.

19.3 VOLATILITY INDICES AND VARIANCE SWAPS

Chapter 1 introduced VIX and Section 18.12 used it; this section defines it. **VIX** is Cboe's volatility index for the S&P 500: the annualized volatility, in percentage points, that the index's

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listed options imply over the next 30 days. It is a construction from the surface, and the construction is Chapter 12's replication result applied once more.

The static portfolio that pays variance. Suppose one wants a payoff equal to the *realized variance* of the underlying over $[0, T]$ (a traded contract defines it as a discrete sum of squared daily log returns, Section 22.2; the continuous-time definition used here is for pricing and hedging). Section 12.8 showed that any payoff can be built from options; the specific answer here is that realized variance is replicated by a static portfolio of out-of-the-money options weighted by $1/K^2$, together with a dynamic position in the underlying. The fair value of that portfolio gives the **fair variance swap rate**:

$$\sigma_{\text{var}}^2 = \frac{2e^{rT}}{T}\left[\int_0^F \frac{P(K)}{K^2}\,dK + \int_F^\infty \frac{C(K)}{K^2}\,dK\right] \tag{19.2}$$

where F is the forward price of the underlying to expiration, and P and C are put and call prices at strike K. VIX is 100 times the square root of (19.2) at 30 days, interpolated in total variance $\sigma^2 T$ between the two listed expirations that bracket 30 days. Cboe computes the same index at other horizons, from nine days (VIX9D) to one year (VIX1Y) and, since April 2023, one day (VIX1D), the reference for Chapter 23's same-day options.

The $1/K^2$ weighting is not a modeling choice; it is what makes the portfolio's payoff proportional to variance rather than to something else. Section 22.3 derives this strip of options, and [TN22] takes up the correction the exchange applies to it in practice. The weighting has a consequence that decides how the index behaves: low strikes receive far more weight than high ones. Because the skew makes those low strikes expensive, a volatility index built this way systematically exceeds the at-the-money implied volatility, by more than four volatility points on Figure 19.1's one-year skew, as Section 22.6 measures. And because real quotes do not run to zero and infinity, Cboe truncates the strip by a mechanical rule, so the published index depends on how far out options happen to be quoted that day.

The instrument behind the formula is the **variance swap**, which pays the difference between realized variance and the fixed rate σ_{var}^2. It became the standard volatility instrument because its payoff is the realized variance whatever price level the moves occurred at, and because its replicating options are held *statically*. The direction of the logic matters here: VIX is not an input to the surface; it is an output of it.

19.4 MODELS THAT FIT THE SMILE

The Black–Scholes–Merton model assumes one volatility, and the market quotes many. Three families of model resolve the contradiction, and each buys something at a price.

Local volatility. Let volatility be a deterministic function of the price and time, $\sigma(S, t)$. Dupire showed that for any arbitrage-free surface, given at every strike and maturity, there is exactly one such function reproducing it. Local volatility therefore fits the smile *perfectly*, by construction, and it preserves completeness: the market has one source of risk, so Section 13.4's replication argument survives and every option is hedgeable.

The cost is its dynamics. A local-volatility model that matches today's surface generally predicts that tomorrow's smile will flatten and shift in ways the market does not display, which matters for anything whose value depends on how the surface moves, such as Section 19.11's delta and the exotic options of Chapter 34.

Stochastic volatility. Let volatility be a second random process, correlated with the price. Heston's model is the standard example, and the negative correlation between price and volatility generates the equity skew directly: it is the leverage explanation of Section 19.9 written as a model.

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The cost is completeness. There are now two sources of risk and one hedging instrument, so by Section 13.4 the market is incomplete, no unique price exists, and options must be used to hedge options. That is not a defect of the model; it is the model reporting a fact about the world that the Black–Scholes–Merton model concealed.

Jumps. Allow the price to move discontinuously. At short maturities, exactly where Figure 19.1 shows the steepest skew, jumps produce fat tails without requiring an implausibly high volatility for the continuous moves. They also break completeness, and for a sharper reason: a jump of unknown size cannot be hedged by any position in the underlying taken before it.

SABR. In interest rate markets the standard is SABR (stochastic alpha, beta, rho), a stochastic-volatility model with a closed-form implied-volatility approximation, which lets a desk fit a smile with four intuitive parameters rather than a numerical procedure. Chapter 29 uses it for swaptions, and Section 29.9 fits one to market data.

No model dominates: each is used where its virtue is the requirement, and exotics desks mostly run a hybrid of the first two, the local-stochastic volatility of Section 21.9, and compare it against the others. Section 19.12 explains why the disagreement between them is itself the number to look at.

19.5 CONSTRAINTS ON A SURFACE

Because the implied density must be a density, a volatility surface cannot take just any shape. Section 11.3 derived the constraints without a model; they now become tests for a surface.

Nonnegative density. From (19.1), $\partial^2 c/\partial K^2 \geq 0$, which is exactly the convexity condition (11.3). A surface that makes call prices nonconvex in the strike implies negative probabilities and admits the butterfly arbitrage of Example 11.2.

Monotonicity. $c(K)$ must be decreasing in K, though by no more than the present value of the strike gap (Section 11.3).

Calendar consistency. For a fixed strike (strictly, a fixed forward moneyness K/F, which is the same thing when there is no carry), total variance $\sigma^2(K,T)\,T$ must be nondecreasing in T. If it were not, a calendar spread would have a negative price for a nonnegative payoff.

These are the conditions any interpolation must respect, and respecting them is harder than it sounds. Fitting a smooth curve through quoted volatilities is not the same as fitting an arbitrage-free surface, and a naive spline through market data routinely produces negative densities somewhere between the quotes. Dealers' pricing systems parameterize the surface instead, most often with Gatheral's stochastic-volatility-inspired (SVI) parameterization, which fits each expiration's total variance with five parameters and is linear in log-strike in the wings; its surface extension, surface SVI (SSVI; Gatheral and Jacquier 2014), comes with explicit conditions under which the whole surface passes the tests above. The linear wings are required: Lee (2004) showed that without arbitrage total implied variance can grow at most linearly in log-strike, $\sigma^2 T$ at most about $2\,|\ln(K/F)|$ far out in either wing. A quadratic smile like Example 19.1's grows faster and must eventually imply an arbitrage; that one does so near a strike of 6, where its density turns negative. In a rising call wing the monotonicity condition is the one to watch, and Example 19.3 turns it into a number a trader can carry.

Example 19.3 How fast a wing may steepen

Take spot at 100, six months to expiration and a 3% rate. The strike enters the price twice, directly and through the implied volatility the smile assigns to it, so differentiating along the smile is a total derivative:

$$\frac{dc}{dK} = \frac{\partial c}{\partial K} + \frac{\partial c}{\partial \sigma}\frac{d\sigma}{dK} = -e^{-rT}N(d_2) + \mathcal{V}\frac{d\sigma}{dK}$$

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The first partial is minus the digital a flat volatility would give (Section 12.8), negative because a higher strike is worth less; the second is vega, positive. A rising wing pushes against the first term, and the call price stops falling in the strike (the call spread turns negative) once

$$\frac{d\sigma}{dK} > \frac{e^{-rT}N(d_2)}{\mathcal{V}}$$

The right-hand side is a hard ceiling on the wing's slope, and it is not constant:

Strike	σ	Vega	Maximum steepening per 5 of strike
110	22.5%	25.77	5.34 volatility points
115	24.0%	22.83	4.45 volatility points
120	26.0%	20.19	3.86 volatility points
125	29.0%	18.60	3.49 volatility points

The ceiling tightens as the strike moves out, because $N(d_2)$ falls faster than vega does, so an extrapolation that steepens the wing as it goes runs the wrong way.

Now breach it. Take a chain quoting 24.0%, 23.2%, and 22.8% at 100, 105, and 110, then an aggressive wing at 25.0% and 31.0% for 115 and 120:

Strike	σ	Call	Call spread
100	24.0%	7.4824	
105	23.2%	5.0715	+2.4109
110	22.8%	3.3237	+1.7478
115	25.0%	2.6563	+0.6673
120	31.0%	3.0442	**−0.3878**

The 115/120 call spread has a payoff that is never negative and a price of −\$0.3878: you are paid \$0.39 to receive something worth between \$0 and \$5.

The chain tolerates $\sigma(120)$ up to 29.27% before the spread turns, so the quoted 31.0% was 1.7 volatility points past the boundary, far too small a discrepancy to see by eye on a plotted smile, and an arbitrage nonetheless. That is why the check is arithmetic rather than visual, and why it is run on the quotes before any curve is fitted through them.

19.6 THE TERM STRUCTURE AND ITS DYNAMICS

The surface has a second dimension, and it moves differently from the first.

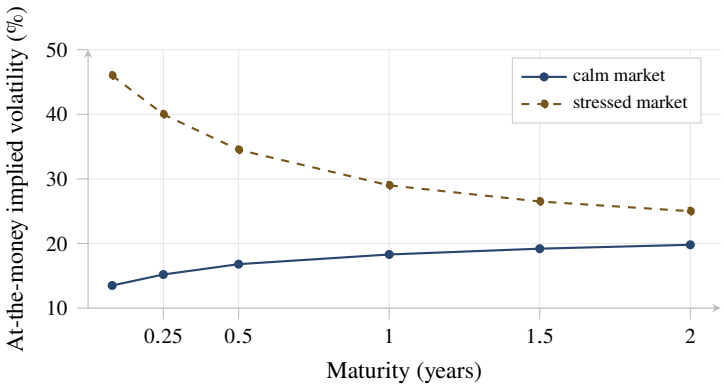


Figure 19.3: At-the-money implied volatility against maturity, in two regimes. Illustrative.

Figure 19.3 shows the characteristic pattern. In calm markets the term structure slopes *upward*: short-dated volatility is low and long-dated volatility is higher. In stressed markets it

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inverts, sometimes violently, with one-month volatility far above two-year.

The mechanism is mean reversion. Volatility is not a random walk. It clusters, as Section 14.12 noted and Section 38.10 models, but it also returns toward a long-run level. A short-dated option prices volatility over a period in which today's level dominates; a long-dated option prices an average over a period in which mean reversion has had time to act. The slope simply reports which side of the long-run level the market is currently on. Two consequences follow for the desk.

Vega buckets are not interchangeable. Section 18.8 said desks compute vega by maturity bucket rather than as one number, and Figure 19.3 is the reason: short maturities move several times as much as long ones. Hedging total vega hedges a parallel shift that does not occur.

Calendar spreads are a term-structure trade. Section 12.5 described the calendar spread as a bet on time decay and on the term structure. Now the second half is precise: selling near-dated and buying far-dated volatility is a bet that short-dated volatility stays low or, when it is high, falls back toward its long-run level, and it loses in exactly the stress episodes that invert the curve.

A scheduled event bends both dimensions. When a single large move is expected on a known date (an earnings release, a court ruling), the distribution of the price just after it can be bimodal; near-the-money options are then worth more relative to the wings than a lognormal implies, and the smile becomes a frown. Only expirations after the date contain the event. If the one just before trades at 25% and one fifteen trading days out at 35%, and the fourteen ordinary days carry 25%, subtracting their variance from the later expiration's total leaves the event day's: $(0.35^2 \times 15 - 0.25^2 \times 14)/252 = 0.0038$. Its square root, 6.2%, is the standard deviation the market prices for the announcement-day move, against $0.25/\sqrt{252} = 1.6\%$ on an ordinary day.

19.7 THE INDEX SMILE AND THE SINGLE-STOCK SMILE

Table 19.2 recorded that an index skews more steeply than its own constituents. That is not an observation about two unrelated markets: the index *is* the constituents, so the two surfaces are linked, and what links them is correlation.

For a portfolio with weights w_i and component volatilities σ_i,

$$\sigma_I^2 = \sum_i w_i^2 \sigma_i^2 + \sum_{i \neq j} w_i w_j \, \rho_{ij} \, \sigma_i \sigma_j \tag{19.3}$$

Everything on the right except the correlations is observable from single-stock option prices, and the left-hand side is observable from index option prices. If all pairs are assumed to share one correlation, that one unknown can be solved for, and the answer is called the **implied correlation**, which Example 19.4 computes.

Example 19.4 Implied correlation and what dispersion trades

Take an index of 50 equally weighted stocks, each with a one-year implied volatility of 30%, and a one-year index implied volatility of 18%. With a common correlation ρ, (19.3) collapses to

$$\sigma_I^2 = \sigma^2 \left[\frac{1}{n} + \rho \left(1 - \frac{1}{n} \right) \right]$$

so $(0.18/0.30)^2 = 0.36 = 0.02 + 0.98\rho$ and $\rho = 34.7\%$.

Where the trade is. Suppose realized correlation over the past year has been 25%. At that correlation the index volatility would be $0.30\sqrt{0.02 + 0.25 \times 0.98} = 15.4\%$, against the 18% the index options actually cost. Selling index volatility and buying the components, the **dispersion trade** of Sections 16.2 and 18.7, collects that gap of 2.6 volatility points, a difference in volatility rather than a 2.6% return, if correlation stays at 25% and the components realize their 30%.

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Why the gap persists. It is Section 19.10 again, in a specific form. Index puts are the instrument institutions buy for portfolio protection, and protection is a demand for correlated downside, so the premium concentrates in the index rather than in the constituents. Implied correlation exceeding realized correlation is the price of that demand.

And why the trade is not free. In a crash component volatilities and correlation rise together, so index volatility rises proportionally more than single-stock volatility, and the long single-stock leg covers only part of the loss. The dispersion seller is short exactly the thing that moves most, which is why the strategy earns steadily and loses in large, infrequent amounts (Example 22.8).

The same mechanism explains the shape difference the table recorded. Correlation is not a constant: it rises when markets fall and falls when they rise. On the downside the index gets hit twice, by higher component volatility and by higher correlation together; on the upside dispersion increases and damps the index move. So the index inherits its constituents' skew and then adds a correlation skew on top of it, and the sum is steeper than any single stock's.

19.8 TRADING THE SMILE

A single surface has three dimensions of shape (level, skew, and curvature), and Section 16.4 already observed that the FX market quotes precisely those three because they are what it trades. The instruments are Chapter 12's, and correlation, which spans two surfaces, is traded through the dispersion trade of Section 19.7.

Level is traded with straddles or, more cleanly, with the variance swaps of Section 19.3.

Skew is traded with the risk reversal of Section 12.6: long an out-of-the-money call, short an out-of-the-money put, or the reverse. A trader who thinks puts are too expensive relative to calls buys the risk reversal, selling the expensive put and buying the cheap call, and must include Section 19.11's delta correction in the hedge.

Curvature is traded with the vega-weighted butterfly, the position the foreign-exchange market's butterfly quote of Section 16.4 prices: a strangle bought against an at-the-money straddle, sized so the two vegas cancel. Buying the wings against the at-the-money straddle is a bet that the market will come to price more extreme outcomes in both directions.

Who is on each side. Most of the participants are the ones Section 16.2 named, and their positions in surface terms are now legible. Protection buyers are structurally long the put wing. Covered-call writers are structurally short the call wing. Structured-product issuers hold the other side of what their products embed: long the barrier puts that buyers of Section 12.12's reverse convertibles and autocallables write, which sit low in the strike dimension, and short the calls inside principal-protected notes. Each of these leaves a persistent mark on a specific region of the surface, and dealers hold whatever the end users leave unmatched. A region expensive relative to its neighbors usually reflects who has been buying it rather than a superior forecast, and Section 19.10 shows that, on average, the buyers overpay.

19.9 WHY THE SMILE EXISTS

Four explanations are offered; all four contribute, and their relative weights are contested.

The distribution really is not lognormal. Section 14.12 gave the reasons: prices jump, and volatility is neither constant nor independent of the price level. A model with jumps or with stochastic volatility produces fatter tails than the lognormal, and fitting the Black–Scholes–Merton model to options priced under such a model generates a smile as an artifact. This explanation is certainly part of the answer, and it explains the FX smile well.

Leverage. A share is a residual claim on a levered balance sheet. When the equity value falls, leverage rises, so the equity becomes more volatile: volatility and price are negatively

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correlated by construction. This produces exactly the asymmetry seen in equities and *not* in currencies, where no such mechanism exists.

Crash risk. The equity skew dates from October 1987, as this section's case study shows. Before that day the market had not priced a 20% one-day decline; afterward it did. On this reading the skew is the price of a risk the lognormal model assigns essentially zero probability to. Rubinstein (1994), who documented the change, called it "crash-o-phobia."

Supply and demand. One-directional demand for low strikes and one-directional supply of high strikes (Section 19.8) will produce the observed shape whatever the distribution.

Case Study 19.1 The day the skew appeared

The four explanations above are hard to separate on cross-sectional data. History separates them rather sharply.

Rubinstein (1994) finds index option implied volatilities before October 1987 close to *flat* across strikes, a smile so mild it was regarded as a minor pricing puzzle rather than a feature, though Bates (1991) finds out-of-the-money puts already expensive in the year before the crash. The Black–Scholes–Merton model, on this evidence, described the market it was written for tolerably well for its first fourteen years.

After October 19, 1987, the skew appears, and it does not go away. Nearly four decades later the pattern of Figure 19.1 is a permanent fixture of every equity index option market in the world.

That timing is difficult to reconcile with the leverage explanation, which describes a mechanism that was equally true in 1986. It is difficult to reconcile with a change in the true distribution, since nothing about the economy changed that Monday. What changed was the market's belief about the probability of a very large decline, revised by a single observation and never revised back.

The supply-and-demand explanation gets support from the same event, which Section 40.2 tells in full: the failure of portfolio insurance destroyed the belief that crash protection could be manufactured by dynamic trading, which turned investors who had made their own puts into buyers of listed ones. Both readings point the same way, and both say the skew is priced crash risk rather than a measured distribution.

Because the four explanations predict a similar shape, the best evidence is the next section's: a skew that merely described the distribution would be fairly priced on average, and it is not.

19.10 THE VOLATILITY RISK PREMIUM

Sections 10.12 and 12.4 asserted a result this chapter owes: implied volatility exceeds subsequently realized volatility, on average, across most markets and most periods (Carr and Wu 2009). That average gap is the volatility risk premium that Section 15.7 named. The magnitude is typically two to four volatility points on equity indices (an implied volatility of 18% against a realized 15%, say), and the gap is larger for low strikes than at the money, so the skew is steeper than the realized distribution justifies. Two readings are offered, and both are partly right.

It is compensation. Selling options means accepting a payoff that is worst precisely when markets fall and investors most need money. It is Chapter 18's short-gamma position, in economic terms. A risk that pays badly in bad states commands a premium, exactly as equities do relative to bills.

It is supply and demand. The buyers of protection described in Section 16.2 are price-insensitive relative to the sellers, who are constrained by capital. The premium is then a rent

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earned by whoever can bear the risk.

Section 12.4's straddle buyer pays this premium, so buying options is a bet that realized will exceed implied, not that the stock will move. Section 12.12's covered-call funds collect it, so their option income earns a genuine expected return, though part of each distribution is the investor's own capital handed back. And selling options earns a risk premium with a short-gamma position, profitable on average and, as Section 18.10's gap example showed, capable of losing more than the whole premium in a single move.

This also settles the question Section 15.7 raised. When an implied volatility of 29.85% meets a historical estimate of 20.4%, there are two readings: the market anticipates something the history does not contain, or buyers are paying the premium. Within a single implied volatility the two components cannot be told apart; averaged over enough observations, the forecast-error component tends to zero, and what remains is the premium.

19.11 HOW THE SMILE MOVES

For a desk, what matters more than the smile at one instant is what happens to it when the underlying moves, and Section 18.10 warned that the answer changes the Greeks. Two reference conventions describe it, and a third completes Derman's (1999) taxonomy.

Sticky strike holds that the implied volatility of a *given strike* stays put as the spot moves. The smile is fixed in strike space, and Black–Scholes deltas are correct as computed.

Sticky delta (or sticky moneyness) holds that the implied volatility of a given *moneyness* stays put, so the whole smile translates with the spot. A fixed strike then acquires a different implied volatility as the market moves.

Sticky implied tree, named for the tree form of local volatility, holds that the local-volatility function of Section 19.4 stays put, so a rise in the spot *lowers* the implied volatility at a fixed strike and the delta falls below the Black–Scholes delta. Derman associated sticky strike with a range-bound index, sticky delta with a trending one, and the implied tree with a market that fears a sharp fall.

Figure 19.4 shows what each convention does to the one-year smile when the spot rises from 100 to 105.

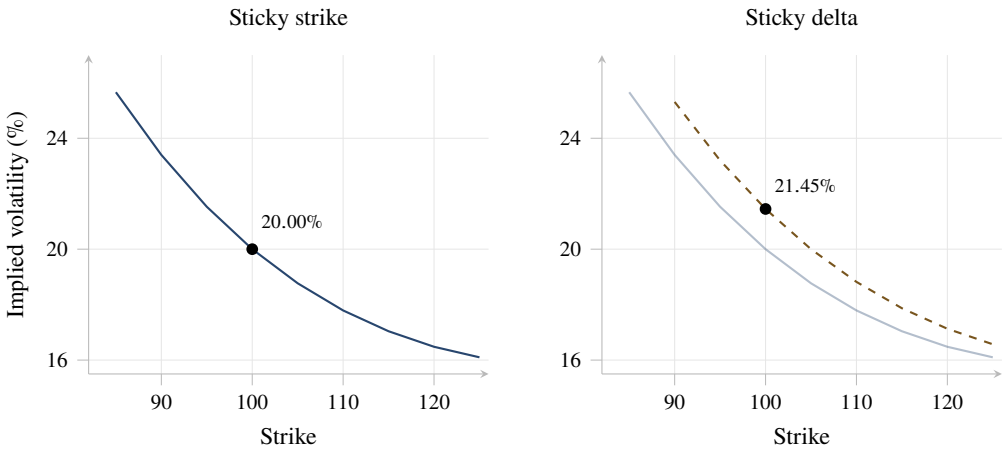


Figure 19.4: The one-year smile after the spot rises from 100 to 105. The marked point is the 100-strike option. Faint curve: the smile before the move.

Under sticky strike nothing happens to the curve, and the 100-strike option keeps its 20.00% volatility. Under sticky delta the curve slides to the right in proportion to the spot, and the 100-strike option, now 5% in the money rather than at the money, picks up the 21.45% that

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used to belong to the 95.2 strike. The same option at the same instant carries volatilities 1.45 volatility points apart under the two conventions, and Example 19.5 shows what that does to the hedge.

Example 19.5 The delta correction

Under sticky delta the spot enters the price twice, directly and through the implied volatility the smile hands the option as it slides, so the sensitivity to the spot is a total derivative:

$$\Delta_{\text{total}} = \frac{dc}{dS} = \frac{\partial c}{\partial S} + \frac{\partial c}{\partial \sigma}\frac{\partial \sigma}{\partial S} = \Delta_{\text{BS}} + \mathcal{V}\frac{\partial \sigma}{\partial S} \tag{19.4}$$

The first partial is the usual delta, computed with the smile held still; the second is vega times the rate at which the sliding smile assigns the option a new volatility. In Chapter 18's expansion, equation (18.1), the delta and vega terms were treated as independent; a sliding smile makes $\Delta\sigma$ move with ΔS, so part of the vega term is first order in the stock and belongs in the delta hedge.

For the one-year skew of Figure 19.1 with $S = 100$ and $r = 0$:

Strike	Type	σ	Δ_{BS}	Correction	Δ_{total}
90	put	0.2340	−0.2853	+0.1256	−0.1596
100	call	0.2000	+0.5398	+0.1092	+0.6490
110	call	0.1779	+0.3275	+0.0683	+0.3958

The corrections run from 0.07 to 0.13 in delta, a 20% error on the at-the-money hedge, or 10,920 shares on a position of 100,000 options. A desk hedging with Black–Scholes deltas while the market behaves in a sticky-delta way has every delta too low: too few shares against a call, too many against a put.

Under an equity skew, volatility falls as strike rises. Sticky delta means the smile moves with spot, so a rise in S lowers the option's moneyness K/S and therefore *raises* its implied volatility. Since vega is positive, the option's value rises by more than delta alone predicts, so every delta in the table moves up: a call's hedge grows and a put's shrinks.

Which convention holds is an empirical question, and for equity indices the measured answer lies outside the first two: when the index falls, at-the-money volatility rises by more than sticky strike implies, though by less than local volatility predicts (Bergomi 2009). The correction for an index therefore has the opposite sign to this table's, and the minimum-variance delta lies below the Black–Scholes delta, as Section 18.10 reported from Hull and White (2017). FX is closer to sticky delta, which suits the delta quoting of Section 16.4. Desks measure the relationship rather than assuming it, and the measured slope enters (19.4) directly.

19.12 THE SURFACE IN PRACTICE

A desk maintains a volatility surface per underlying, and it is the single most important piece of market data an option business owns.

What one actually looks like. Table 19.3 is an S&P 500 surface in the form a desk reads it: implied volatility in percent, strikes across as a percentage of spot, maturities down the side. Its wings are milder than Figure 19.1's.

Every row falls from left to right through the money, which is the skew of Section 19.1. A one-month put struck 20% below spot is marked at 30.0% against 17.5% at the money; the same strike two years out is 24.2% against 20.8%. Measured between the 90 and 110 strikes the slope runs 3.75 volatility points per 10% of spot at one month, 1.70 at one year and 1.15 at two, so two-thirds of the skew is gone by the two-year point. Measured in Section 19.1's standardized coordinate instead, the same rows give 1.89 volatility points per standard deviation at one month against 3.37 at two years: the skew *steepens*. Read down the at-the-money column and the second dimension appears: 17.5% rising to 20.8%, the calm-market term structure of Section 19.6.

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Table 19.3: Implied volatility (%) by strike, as a percentage of spot, and by maturity. A calm market; illustrative, not market data.

	Strike (% of spot)						
Maturity	**80**	**90**	**95**	**100**	**105**	**110**	**120**
1 month	30.0	22.4	19.6	17.5	15.8	14.9	15.6
3 months	27.5	21.9	19.9	18.5	17.1	16.0	14.8
6 months	26.0	21.7	20.3	19.3	18.2	17.2	15.6
1 year	25.0	21.9	20.8	20.0	19.3	18.5	17.2
2 years	24.2	22.0	21.3	20.8	20.2	19.7	18.7

Two features are worth pointing out because they look like errors and are not. Total variance rises down every column: at the 110 strike $\sigma^2 T$ runs from 0.0019 at one month to 0.0776 at two years. That is Section 19.5's calendar condition holding, and a column in which it failed anywhere would be a calendar-spread arbitrage rather than a steep surface. And the top right corner turns back up (the one-month 120 strike at 15.6% against the 110's 14.9%), because no one sells a far out-of-the-money one-month call at the price a straight continuation of the skew would imply.

It is built from quotes, not from a model. Each expiration's forward is read first from put–call parity near the money (Section 11.6), and time runs on Section 15.8's trading-day clock, refined to Chapter 23's business time, which weights each day by its expected variance; both matter most at the short end, where Section 23.9 prices the same-day smile this way. Liquid strikes and maturities are then marked to the market; everything else is interpolated subject to the constraints of Section 19.5.

It prices the illiquid from the liquid. Section 15.8's practical rule was to use each option's own implied volatility. The surface is what makes that possible for an option that does not trade: locate it in strike–maturity space, interpolate, price. *This is what the Black–Scholes–Merton formula is actually for.* Example 19.6 carries it through on Table 19.3.

Example 19.6 Pricing an option that does not trade

A client asks for a nine-month put struck at 4,600 on an index trading at 5,000. No line of Table 19.3 is that option: the strike is 92% of spot and the maturity falls between two quoted rows. It sits inside the rectangle formed by the 90 and 95 strikes and the six-month and one-year maturities, and those four corners are what price it.

Interpolate across the strikes first, linearly, at each of the two maturities:

$$\sigma(92, 0.5) = 21.7 + 0.4 \times (20.3 - 21.7) = 21.14\%$$
$$\sigma(92, 1) = 21.9 + 0.4 \times (20.8 - 21.9) = 21.46\%$$

Then move down the maturities, but not in volatility. Interpolate in total variance $\sigma^2 T$, the quantity Section 19.5 requires to be nondecreasing:

$$w(0.5) = 0.2114^2 \times 0.5 = 0.02234, \qquad w(1) = 0.2146^2 = 0.04605$$

At nine months $w = 0.03420$, so $\sigma = \sqrt{0.03420/0.75} = 21.35\%$. Interpolating the two volatilities directly would have given 21.30%. Five hundredths of a point is nothing here, but only the variance rule always respects the calendar bounds the two quoted maturities impose.

With $r = 4\%$ and a dividend yield of 1.5%, the Black–Scholes–Merton formula at 21.35% gives a put worth 157.9 index points.

Interpolated down the 100 column the same way, the nine-month at-the-money volatility is 19.77%; priced at that single volatility, the same put is worth 136.2. A desk doing so would have sold the option 13.8% below its market value, and the missing 21.7 index points are the skew: the

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> option's own strike was never allowed to enter its price.

It generates the risk report. Chapter 18's vega is computed bucket by bucket against this surface, and the buckets are chosen where the surface actually moves independently.

Its failures are exotic risk. A vanilla option is priced by reading the surface. An exotic is not: Chapter 34 shows that barrier and path-dependent payoffs depend on how the surface *evolves*, which no static surface specifies. Two models that fit today's surface perfectly can price the same barrier option far apart. Section 21.11 measures the gap at nearly half the price for a reverse knock-out (a call knocked out by a barrier above its strike, so it dies where it is in the money), and at about one percent for a down-and-out call struck at the money (knocked out by a fall through a barrier below). That gap, not the vanilla price, is the model risk that Section 13.4's completeness discussion predicted, and its size depends on where the payoff sits relative to the region the surface leaves unconstrained.

SUMMARY

Implied volatility is not constant across strikes. Equities show a skew (low strikes expensive, high strikes cheap) that has persisted since October 1987; currencies show a roughly symmetric smile whose tilt changes sign with sentiment. The skew flattens with maturity, mostly because a fixed strike gap spans fewer standard deviations at longer horizons; persistent volatility keeps the long-dated skew from vanishing as independent shocks alone would let it. A skew number means nothing without the moneyness convention it was measured in.

By Breeden–Litzenberger, $g(K) = e^{rT}\partial^2 c/\partial K^2$, so a complete set of option prices contains the market's entire risk-neutral distribution. Applied to a skew it gives a fat left tail and a thin right one, with the market assigning a large decline several times the probability the lognormal does; run on a quoted chain rather than a fitted curve, butterflies give the density and call spreads give the tails. Beyond the lowest traded strike, extrapolation decides the tail's shape.

Because the density must be nonnegative, Section 11.3's convexity condition becomes a constraint on any surface, alongside monotonicity in strike and nondecreasing total variance in maturity. Fitting a smooth curve is not the same as fitting an arbitrage-free surface: a wing can steepen only so far across a range of strikes before the quotes are themselves an arbitrage.

A volatility index is an output of the surface: a $1/K^2$-weighted strip of out-of-the-money options prices variance, and the skew lifts it above at-the-money volatility. Models that fit the smile trade one virtue against another: local volatility fits today's surface exactly and keeps the market complete but mispredicts how it moves, while stochastic volatility and jumps capture the dynamics and short-dated tails at the cost of completeness.

Four explanations contribute: a distribution that is not lognormal, leverage, crash risk priced since 1987, and one-directional demand. Leverage and crash risk give equities a direction that currencies lack; the timing of 1987 and the index's steeper skew point to priced crash risk and one-directional demand.

The term structure moves on its own logic: at-the-money volatility slopes upward in calm markets and inverts in stressed ones, which is why vega is bucketed by maturity. Component and index volatilities together imply a correlation, and selling it against realized correlation is the dispersion trade. Implied volatility on equity indices also exceeds subsequently realized volatility by a few points on average, so buying a straddle is a bet that realized will exceed implied rather than a bet that the market will move.

How the smile moves determines the hedge. The total delta is $\Delta_{\mathrm{BS}} + \mathcal{V}\,\partial\sigma/\partial S$, a correction too large to ignore; sticky delta makes it positive, while on equity indices, where fixed-strike volatility falls as the index rises, it is negative, so desks measure it rather than assume it. The surface is built from quotes subject to those constraints, prices an option that does not trade

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consistently with ones that do, and generates Chapter 18's vega buckets. Its limitation is that it says nothing about its own evolution, which is where exotic model risk lives.

FURTHER READING

Breeden and Litzenberger (1978) is the original statement of (19.1) and is short. Derman (1999) sets out the sticky-strike, sticky-delta and sticky-implied-tree regimes of Section 19.11, and Bergomi (2009) measures how index smiles actually move against them. Daglish, Hull and Suo (2007) derive the no-arbitrage condition a moving surface must satisfy and test the sticky rules against it and against S&P 500 data; a skewed surface is consistent with neither rule in its basic form. Gatheral's *The Volatility Surface* (2006) is the standard reference for arbitrage-free parameterizations and the constraints of Section 19.5; Gatheral and Jacquier (2014) give SSVI and Lee (2004) the wing bound. On the volatility risk premium, Carr and Wu (2009) measure it from synthetic variance swaps across indices and single stocks, and Bollerslev, Tauchen and Zhou (2009) relate it to expected stock returns; Chapters 21 and 22 return to it when volatility becomes a traded asset.

PRACTICE QUESTIONS

19.1 Explain why equity options show a skew while currency options show a roughly symmetric smile, giving the structural reason in each case.

19.2 Using the three-month and six-month rows of Table 19.3, compute the slope between strikes 90 and 110 per 10% of spot and per standard deviation of standardized moneyness (use each row's at-the-money volatility), and explain why the two disagree about whether the skew flattens.

19.3 State the Breeden–Litzenberger result and explain, without algebra, why the second derivative of the call price with respect to strike is a probability density.

19.4 Using the table in Example 19.1, state by what factor the market's probability of a 30% decline exceeds the lognormal's, and explain what feature of the smile produces it.

19.5 Using Example 19.3, state the largest rise in implied volatility the 115/120 spread can carry, describe the arbitrage that appears beyond it, and explain why the permitted slope shrinks as strikes move further out of the money.

19.6 An at-the-money option has vega 40 and Black–Scholes delta 0.54. Under sticky delta with $\partial\sigma/\partial S = 0.0015$, compute the corrected delta and the error in shares on a 50,000-option position, and state which convention gives the larger hedge for a call under an equity skew.

19.7 Explain why the volatility risk premium implies that a straddle buyer is not simply betting the market will move.

19.8 In Example 19.2 the bucket probabilities and the two tail probabilities sum to exactly 100%. Explain why this had to happen and what it would mean if it did not; then replace the \$50 call's midquote of 1.475 with 1.050, and say which butterfly turns negative and what arbitrage it offers.

19.9 An index of 100 equally weighted stocks has component implied volatilities of 25% and an index implied volatility of 16%. Compute the implied correlation, and explain why index skew is steeper than single-stock skew.

19.10 Explain why the volatility term structure inverts in a crisis, and why this makes total vega a poor summary of an option book's exposure.

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FURTHER QUESTIONS

19.11 Example 19.1 notes that its quadratic smile, extended below the observable range, implies a density that turns negative below a strike of about 6 and totals 1.06 when integrated upward from there. Explain why this is not a numerical error to be corrected, describe two different extrapolation choices and how each would distribute the probability below the lowest observable strike (a total the chapter shows is fixed), and state what a risk manager should conclude about tail estimates derived this way.

19.12 Section 19.9 gives four explanations for the skew. Design an empirical test that would distinguish the leverage explanation from the supply-and-demand explanation, stating what data you would need and what each hypothesis predicts.

19.13 Section 19.10 states that the volatility risk premium is larger at low strikes than at the money. Explain what this implies about the profitability of selling puts versus selling straddles, and why the strategy with the higher average return is not obviously the better one.

19.14 Show that under sticky delta the at-the-money-forward implied volatility does not change when the spot moves, even though the correction in (19.4) for the option struck there today does not vanish (Example 19.5), and explain why this makes the at-the-money-forward straddle of Section 11.4 a natural reference point for a desk.

19.15 Section 19.12 claims two models fitting today's surface perfectly can price the same barrier option far apart, and that the size of the gap depends on the payoff. Explain what information a barrier price requires that a vanilla price does not, connect this to Section 13.4's completeness argument, and explain why a reverse knock-out should be far more exposed to the modeling choice than a down-and-out struck at the money.

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